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Report No: PAD3760

INTERNATIONAL BANK FOR RECONSTRUCTION AND DEVELOPMENT

PROJECT APPRAISAL DOCUMENT

ON A

PROPOSED GRANT

IN THE AMOUNT OF
US\$16,365,000

TO

JAMAICA

FOR A

CATASTROPHE BOND FOR INCREASED FINANCIAL RESILIENCE TO NATURAL
DISASTERS AND CLIMATE SHOCKS PROJECT

June 24, 2021

Finance, Competitiveness and Innovation Global Practice
Latin America and Caribbean Region

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CURRENCY EQUIVALENTS

(Exchange Rate Effective June 3, 2021)

Currency Unit = Jamaican Dollar (J\$)

146.85 J\$ = US\$1

FISCAL YEAR

April 1 - March 31

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ABBREVIATIONS AND ACRONYMS

CCRIF	Caribbean Catastrophe Risk Insurance Facility
CCRIF SPC	Caribbean Catastrophe Risk Insurance Facility Segregated Portfolio Company
CAT bond	Catastrophe Bond
CPF	Country Partnership Framework
DPF	Development Policy Financing
DRF	Disaster Risk Financing
ESCP	Environmental and Social Commitment Plan
ESMF	Environmental and Social Management Framework
ESMS	Environmental and Social Management System
E&S	Environmental and Social
FCDO	Foreign, Commonwealth and Development Office
FCI	Finance Competitiveness and Innovation
GDP	Gross Domestic Product
GOJ	Government of Jamaica
GRIF	Global Risk Financing Facility
GRM	Grievance Redress Mechanism
GRS	Grievance Redress Service
IADB	Inter-American Development Bank
IBRD	International Bank for Reconstruction and Development
IMF	International Monetary Fund
ILS	Insurance-Linked Securities
IPF	Investment Project Financing
LCR	Latin America and Caribbean
LMP	Labor Management Plan
MDTF	Multi-donor Trust Fund
MOFPS	Ministry of Finance and the Public Service
NCRA	Natural Resources Conservation Authority
NDF	National Disaster Fund
NDMO	National Disaster Management Office
NNDRF	National Natural Disaster Reserve Fund
NPV	Net Present Value
PDO	Project Development Objective
SEP	Stakeholder Engagement Plan
RTA	Risk Transfer Agreement
SPV	Special Purpose Vehicle
TA	Technical Assistance
TRE	World Bank Treasury
TC-W	Tropical Cyclone-Wind
UK	United Kingdom
WB	World Bank
WBG	World Bank Group

TABLE OF CONTENTS

DATASHEET	Error! Bookmark not defined.
1. STRATEGIC CONTEXT	6
A. Country Context.....	6
B. Sectoral and Institutional Context	8
C. Relevance to Higher Level Objectives.....	11
2. PROJECT DESCRIPTION.....	13
A. Project Development Objective	13
B. Project Components	13
C. Project Beneficiaries	14
D. Results Chain	14
E. Rationale for Bank Involvement and Role of Partners	15
F. Lessons Learned and Reflected in the Project Design	18
3. IMPLEMENTATION ARRANGEMENTS	19
A. Institutional and Implementation Arrangements.....	19
B. Results Monitoring and Evaluation Arrangements.....	20
C. Sustainability.....	20
4. PROJECT APPRAISAL SUMMARY	21
A. Technical, Economic and Financial Analysis.....	21
B. Fiduciary.....	26
C. Legal Operational Policies.....	27
D. Environmental and Social	27
5. GRIEVANCE REDRESS SERVICES	29
6. KEY RISKS	29
7. RESULTS FRAMEWORK AND MONITORING	31
ANNEX 1: Implementation Arrangements and Support Plan	33
ANNEX 2: Catastrophe Bonds	38
ANNEX 3: Risk Layering Approach	40
ANNEX 4: 360 Degree Resilience Framework.....	41



DATASHEET

BASIC INFORMATION

Country(ies)	Project Name	
Jamaica	Jamaica Catastrophe Bond for increased Financial Resilience to Natural Disasters and Climate Shocks	
Project ID	Financing Instrument	Environmental and Social Risk Classification
P173012	Investment Project Financing	Low

Financing & Implementation Modalities

☐ Multiphase Programmatic Approach (MPA)	☐ Contingent Emergency Response Component (CERC)
☐ Series of Projects (SOP)	☐ Fragile State(s)
☐ Performance-Based Conditions (PBCs)	☐ Small State(s)
☐ Financial Intermediaries (FI)	☐ Fragile within a non-fragile Country
☐ Project-Based Guarantee	☐ Conflict
☐ Deferred Drawdown	☐ Responding to Natural or Man-made Disaster
☐ Alternate Procurement Arrangements (APA)	☐ Hands-on Enhanced Implementation Support (HEIS)

Expected Approval Date	Expected Closing Date
30-Jun-2021	21-Jun-2024
Bank/IFC Collaboration	
No	

Proposed Development Objective(s)

To expand Jamaica's financial protection against losses arising from severe tropical cyclones-wind.

**Components**

Component Name	Cost (US\$, millions)
Development and implementation of risk transfer transactions against tropical cyclones-wind effects	16,515,000.00

Organizations

Borrower:	Jamaica
Implementing Agency:	Ministry of Finance and the Public Service

PROJECT FINANCING DATA (US\$, Millions)**SUMMARY**

Total Project Cost	16.52
Total Financing	16.52
of which IBRD/IDA	0.00
Financing Gap	0.00

DETAILS**Non-World Bank Group Financing**

Counterpart Funding	0.15
Borrower/Recipient	0.15
Trust Funds	16.37
Global Facility for Disaster Reduction and Recovery	16.37

Expected Disbursements (in US\$, Millions)

WB Fiscal Year	2021	2022
Annual	16.37	0.00



Cumulative	16.37	16.37
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INSTITUTIONAL DATA

Practice Area (Lead)

Contributing Practice Areas

Finance, Competitiveness and Innovation

SYSTEMATIC OPERATIONS RISK-RATING TOOL (SORT)

Risk Category

Rating

1. Political and Governance	● Low
2. Macroeconomic	● Moderate
3. Sector Strategies and Policies	● Low
4. Technical Design of Project or Program	● Moderate
5. Institutional Capacity for Implementation and Sustainability	● Low
6. Fiduciary	● Low
7. Environment and Social	● Low
8. Stakeholders	● Low
9. Other	
10. Overall	● Moderate

COMPLIANCE

Policy

Does the project depart from the CPF in content or in other significant respects?

☐ Yes ☒ No



Does the project require any waivers of Bank policies?

☒ Yes ☐ No

Have these been approved by Bank management?

☒ Yes ☐ No

Is approval for any policy waiver sought from the Board?

☐ Yes ☒ No

Environmental and Social Standards Relevance Given its Context at the Time of Appraisal

E & S Standards	Relevance
Assessment and Management of Environmental and Social Risks and Impacts	Relevant
Stakeholder Engagement and Information Disclosure	Relevant
Labor and Working Conditions	Relevant
Resource Efficiency and Pollution Prevention and Management	Not Currently Relevant
Community Health and Safety	Not Currently Relevant
Land Acquisition, Restrictions on Land Use and Involuntary Resettlement	Not Currently Relevant
Biodiversity Conservation and Sustainable Management of Living Natural Resources	Not Currently Relevant
Indigenous Peoples/Sub-Saharan African Historically Underserved Traditional Local Communities	Not Currently Relevant
Cultural Heritage	Not Currently Relevant
Financial Intermediaries	Not Currently Relevant

NOTE: For further information regarding the World Bank's due diligence assessment of the Project's potential environmental and social risks and impacts, please refer to the Project's Appraisal Environmental and Social Review Summary (ESRS).



Legal Covenants

Conditions

1. STRATEGIC CONTEXT

A. Country Context

- 1. Jamaica is highly exposed to natural disasters that pose substantial macroeconomic and development risks.** Jamaica is the third most exposed country in the world to multiple hazards, with over 96 percent of the country's gross domestic product (GDP) and population at risk from two or more hazards. Jamaica is highly exposed to tropical cyclones, earthquakes, droughts, floods, and landslides, which occur frequently and at varying intensities.¹ These events pose a significant threat to Jamaica's macroeconomic outlook as underlined by the International Monetary Fund (IMF).² Between 1993 and 2003, 26 natural disasters have resulted in total losses of US\$2.22 billion, or 1.5 percent of Jamaica's average annual GDP over the period, the same amount as the total GDP growth during the period.³ After Hurricane Dean (2007) and Tropical Storm Gustav (2008), Jamaica's inflation rate peaked at over 20 percent. The increase in inflation closely mirrors the rate of change in the debt-to-GDP ratio over the past 15 years. The World Bank (WB) estimates that, under current temperatures, Jamaica's contingent liabilities related to tropical cyclones and floods average US\$121 million annually (J\$17 billion) or 0.8 percent of 2020 GDP. This is equivalent to 1.9 percent of total government expenditures in 2020. Contingent liabilities for events occurring one-in-50 years to one-in-100 years range between 7.34 and 12.0 percent of Jamaica's 2020 GDP. Natural disasters continue to increase Jamaica's sovereign debt level, prompting the need for loans and/or tax increases to finance unplanned post-disaster expenditures. Natural disasters also have important welfare impacts. The average damaging hurricane reduces Jamaican per capita consumption by approximately 1.1 percent; more destructive events can cause reductions multiple times this amount.⁴
- 2. After decades of chronic macroeconomic imbalances, the Government of Jamaica (GOJ) in 2013, with the support of international financial institutions, embarked on a program aimed at stabilising the economy, reducing debt, and fuelling growth.** The program, underpinned by an IMF Stand-By Arrangement (SBA), enjoyed broad public support and was a remarkable success. The GOJ sustained annual

¹ High exposure is attributed to the country's location in the Atlantic Hurricane Belt, the geophysical orientation of its low-lying coastal zones, and its mountainous topography. The Jamaican territory is also crossed by five major fault lines, including the Plantain Garden Fault Zone, which triggered the 2010 Haitian earthquake.

² International Monetary Fund, "2014 Article IV Consultation Fourth Review Under the Extended Fund Facility and Request for Modification of Performance Criteria," IMF Country Report no. 14/169, June 2014.

³ World Bank, 2018. Advancing Disaster Risk Finance in Jamaica. Average annual GDP growth for the 1993-2003 period was 1.5 percent.

⁴ Henry M, Spencer N, Strobl E (2019) The impact of tropical storms on households: evidence from Panel data on consumption. Oxf. Bull Econ Stat 82:1–22. <https://doi.org/10.1111/obes.12328>



primary surpluses above 7 percent of GDP under the program, which, coupled with prudent liability management, led to a 50 percentage points decline in public debt to 94 percent of GDP by March 2020. Low and stable inflation in the target range of 4-6 percent became the norm. Unemployment fell to its lowest levels and reserves were comfortably above 5 months of imports. The favorable turnaround in the country's macroeconomic situation was also reflected in the local stock exchange, which was rated as the world's best performing stock index in 2018. In 2019, rating agencies Standard and Poor's Global Ratings, Fitch Ratings, and Moody's Corporation upgraded the country's sovereign rating to B+, B+, and B2, respectively. The Government's successful effort to consolidate its fiscal balances spans two political administrations and remains anchored in a social partnership (the Economic Program Oversight Committee) involving stakeholders from public, private, and civil society. The steadfast implementation of sound policy reforms strengthened the economy and allowed for more broad-based reforms to enhance the country's ability to withstand shocks.

3. **Within six months of the completion of the IMF SBA, the country is confronting a severe economic crisis occasioned by the COVID-19 pandemic.** While the health impact of the pandemic in Jamaica has not been as severe as in some countries, the associated socio-economic impact has been significant, pushing the economy into a deep recession. Jamaica's real GDP contracted by 10.2 percent in 2020, largely due to the impact of COVID-19 on tourism which accounts for over a third of Jamaica's GDP. The government took early and aggressive measures starting in March 2020 to prevent the spread of infection, including cancelling all major public and private gatherings, closing schools, and quarantining entire communities. Curfews across the island remain in place, although the closure of the borders to incoming tourists was lifted on June 1, 2020 for returning Jamaican citizens and on June 15, 2020 for non-citizens. The GOJ also instituted the COVID-19 Allocation of Resources for Employees (CARE) Programme to protect the poor and vulnerable who lost jobs and livelihoods, as well as several initiatives to support businesses to ensure a rapid and sustainable recovery.
4. **Poverty in Jamaica has declined in recent years, but the impact of COVID-19 is likely to cause some reversal.** The poverty rate in 2018 was 12.6 percent, the lowest recorded in 10 years. Nevertheless, poverty reduction has not been continuous, and recorded significant volatility between 2012 and 2018, underscoring the population's vulnerability to short-term shocks. The shock to the economy resulting from the COVID-19 pandemic is expected to affect the welfare of households through reductions in labor income, which if unmitigated could push at least 400,000 Jamaicans into poverty.⁵ Impacts are likely to be uneven across the population, which could exacerbate inequality. Notably, unemployment rose by 5.3 percentage points to 12.6 percent between January and July 2020, but has subsequently declined (8.9 percent in January 2021, as per latest available data).
5. **Climate change is increasing Jamaica's vulnerability to natural disasters, further compromising hard achieved macroeconomic stability.** Climate change models predict Jamaica could be impacted by an increased frequency and/or severity of catastrophic natural disaster events because of heightened surface temperatures and global sea level rise. The 2020 Atlantic hurricane season was the most active and the fifth costliest Atlantic hurricane season on record. In addition, it was the fifth consecutive above-average

⁵ Juan Montecino and Jake Johnson, "Update on the Jamaican Economy," CEPR Reports and Issue Briefs 2012-15, 2012.



season from 2016 onward. Although Jamaica was not directly hit by any tropical cyclone during the season, the island was severely impacted by rainfall-induced flooding and landslides brought on by Delta, Zeta, and Eta during October and November 2020. Expectations of above normal activity also extend to the 2021 season.⁶ Increased exposure to natural disasters presents a challenge to Jamaica's hard-won macroeconomic stability gains, which are already being affected by the COVID-19 pandemic. Given the underlying urgency, the GOJ has embarked on the implementation of a comprehensive disaster risk financing (DRF) strategy.

B. Sectoral and Institutional Context

6. **The GOJ, with the support of the WB, has embarked upon a comprehensive strategy toward developing 360-degree resilience to natural disasters and climate shocks.** The reforms cover financial, physical, and social resilience and are being supported by the WB through various instruments, including Development Policy Financing, Investment Policy Financing (IPF), and technical assistance (TA). The reforms are in varying stages of completion (see Annex 4 for the 360-degree resilience framework for a list of some of the key reforms underway).
7. **The National Public Financial Management Policy Framework and the National DRF Policy underpin the GOJ financial resilience aspects of the 360-resilience approach.** The National Public Financial Management Policy Framework for Natural Disaster Risk Financing was approved by Jamaica's Cabinet in November 2018. A National Disaster Risk Financing Policy, drawing from the framework, is expected to be submitted to Parliament by end November 2021. The policy framework outlines the country's vision and details the enabling environment needed to ensure that adequate resources are available to address ex post financing requirements through a variety of instruments. The policy framework considers cost-effectiveness, timeliness, and sound administrative arrangements for reducing the fiscal impact of natural disasters by proposing a risk-layering strategy that combines risk retention and risk transfer instruments. Several efforts are ongoing to implement the policy framework, including TA from the WB.
8. **While the GOJ has been improving its financial protection, existing financial instruments are insufficient and leave Jamaica exposed to losses resulting from natural disasters.** The GOJ has three main ex-ante financial instruments available to finance public expenditures post-disaster. These are: (i) a Contingencies Fund for Natural Disasters (US\$31 million)⁷; (ii) a Parametric Contingent Credit line (US\$285 million) from the Inter-American Development Bank (IADB) that can be activated in case of earthquakes (measured in the Mercalli scale) and tropical cyclones (measured in terms of precipitation and wind intensity) and after

⁶ On December 9, 2020, the Tropical Storm Risk (TSR) issued an extended range forecast for the 2021 season, predicting activity that is above normal. In this report, the organization predicts 16 named storms, 7 hurricanes and 3 major hurricanes. The main factor behind their prediction is the expected development of a weak *La Niña* by the third quarter of 2021. TSR is a venture developed from a UK government-supported project to predict and map tropical storm activity worldwide.

⁷ In March 2019 GOJ recognized the need to increase its budget for frequent, low-intensity events, allocating an additional US\$15 million to the Contingencies Fund. It is expected that a Sub-account of the Contingencies Fund or a Subsidiary Consolidated Fund will become the country's National Natural Disaster Reserve Fund (NNDRF), with an additional annual allocation of approximately US\$4 million a year for the next three years. This allocation was enabled by a resolution by the Parliament in March 2019, to lift the maximum limits of the Contingencies Fund. In February 2021, the GOJ indicated a balance of JDM4.6 billion, equivalent to US\$31 million.



a certain number of people are affected; and (iii) Caribbean Catastrophe Risk Insurance Facility (CCRIF) Segregated Portfolio Company (SPC)⁸ insurance policies. There is also a National Disaster Fund which is a small fund managed by the Ministry of Local Government and Rural Development which received small annual budget allocations (US\$0.35 million was allocated for fiscal year 2021/2022)⁹. This fund is used for smaller non-national level disasters. Analysis conducted by the WB shows that for tropical cyclone (wind intensity events, TC-W) related post-disasters emergency expenditure only, the financing gap ranges from zero, for small and very frequent events, to massive gaps related to very infrequent events (with frequency of less than one in hundred years or less) which should not be financed with ex-ante instruments¹⁰. However, there is an important financing gap for severe events occurring one in 20 to one in 100 years which may be transferred. Section 4.A of this document on “Technical, Economic and Financial Analysis” presents an evaluation of nine financing strategies, including seven catastrophe (CAT) bond prototypes. The analysis must be redone during the placing process to evaluate the best option for Jamaica.

9. **The WB has been working with the GOJ to develop a CAT bond, the first for a Caribbean state, to reduce the identified financing gap at a reasonable cost and to provide risk diversification benefits.** The Financial Solutions for Climate and Natural Disaster Risks Programs in the Caribbean (P168156), a Trust-funded programmatic advisory services project, provides TA to the GOJ to increase their financial protection against disasters. The proposed CAT bond, an innovative risk transfer instrument (see Annex 2), was identified as complementary to the existing financial instruments that will help to reduce the financing gap. The CAT bond, intermediated by the WB, will complement and leverage the other instruments mentioned (NDF and Contingencies Fund, the CCRIF, and the IADB contingent line of credit).¹¹ Table 1 demonstrates how the CAT bond fits into Jamaica’s broader DRF strategy. The CAT bond will be complementary to other activities and incentivize reforms targeting financial preparedness. This is consistent with the risk layering strategy that is being pursued by the GOJ, as shown in Annex 3. Furthermore, as both the CCRIF SPC and the IADB Contingent Credit are also parametric instruments (i.e. coverage is triggered by a parameter), the addition of the CAT bond with a simplified trigger will diversify the basis risk of the Jamaica disaster risk financing strategy as a whole.¹²

⁸ On June 1, 2021 the CCRIF insurance coverage was kept at about previous year level: US\$ 238.03 million. The current coverage is split as follows: Tropical Cyclone=US\$ 81.63 million, Earthquake=US\$ 125M, Excess Rain=US\$ 31.4 million.

⁹ The fiscal year in Jamaica is April to March.

¹⁰ Financing gap is the difference between contingent liabilities and resources available from DRF instruments.

¹¹ The risk transfer transaction intermediated by a World Bank-issued CAT bond will be referred as the CAT bond throughout the document.

¹² Basis risk in parametric insurance arises when the parameter measurements do not match the insured’s actual losses.



Table 1: Jamaica's portfolio of instruments

	Amount/ Premium (US\$ million)	Coverage limit (US\$ million)	Low risk, high frequency	Moderate risk, moderate frequency	High risk, low frequency
Contingencies Fund for Natural Disasters	---	31.0	✓		
CCRIF (TC, EQ, and XSR)	7.1 per year	238.03		✓	✓
IADB parametric contingent line of credit ^(a)	See note (a)	285.0		✓	
Risk transfer via WB issued CAT bond	6.62 ^(b) per year	175-215 (193= mid point) ^(c)			✓

(a) IADB conditions: tenor 25 years; grace period= 2 years; interest rate= 1.7185 percent. Access to loan based on hard parameters related to "affected population" by earthquake and tropical cyclone (Wind and/or excess of rain).

(b) Assumes a total premium over three years, paid by this IPF and a USAID grant of US\$ 5 million.

(c) The CAT bond coverage range, and mid-point, is indicative (final amount depends on market conditions) and it is specific to the structure already decided by the GOJ, i.e. Modified Prototype 1 [AP=2.42%, EL=1.48%, EP=0.74%].

10. **Financial resources provided by the proposed CAT bond will enhance GOJ ability to cover emergency costs.** One of the major advantages of CAT bonds is that they are specifically designed to provide a relatively quick payout in the event of a predetermined event, so the GOJ would have access to additional financial resources to attend to the initial expenditures (emergency or relief), without having to divert resources planned for other social and development purposes.¹³ Emergency losses in Jamaica, for both earthquake and tropical cyclone, are estimated by the GOJ at 9 percent of reconstruction losses. CAT bond payouts can be allocated to the general budget or to one of the existing disaster risk funds. The GOJ has advised that it intends to create a National Natural Disaster Reserve Fund (NNDRF), to which it will allocate resources from existing DRF instruments. NNDRF it is expected to be created by October 2021. The GOJ will receive TA from the WB to building capacity for effective management of the fund through the enhancement of transparency and accountability in the use of resources, inter alia.

11. **The GOJ fiscal consolidation efforts have gathered donor support for this Project, the first IPF that supports CAT bond issuance.** The GOJ fiscal consolidation efforts have resulted in the international donor community's positive response to support the GOJ's DRF agenda, including by way of the Global Risk Financing Facility (GRiF) Grant to support the payment of premiums for this proposed CAT bond. GRiF provides recipient executed trust funds (RETFs), which above a certain size are processed under IPF policies, to finance the cost-sharing of market-based risk transfer solutions. While the WB has supported other risk transfer solutions via IPFs and has issued CAT bonds for the benefit of countries issued with the sponsor own resources, this is the first IPF that supports a CAT bond issuance.

¹³ Under the proposed structures, payouts in case of insured events would occur within three to five weeks.



12. **The proposed CAT bond will provide insurance coverage against emergency losses arising from tropical cyclones-wind (TC-W).** The WB hired the risk modeling firm AIR Worldwide to provide a probabilistic CAT model for economic losses arising from *earthquake* and *TC-W* and a series of CAT bond prototypes. AIR does not have a CAT model for the peril of *excess rain*. Modelled losses from TC-W are larger than modelled losses from earthquakes. The current financing strategy shows a bigger gap for TC-W than for earthquakes, as the GOJ has bought almost all available earthquake coverage under CCRIF SPC but the purchase for TC-W is limited.¹⁴ Based on this result, and the agreement of the GOJ, the CAT bond prototypes have been designed to cover TC-W emergency losses. Payouts will be triggered based on the central pressure of the tropical cyclone as reported by the U.S. National Hurricane Centre.
13. **CAT bonds present important advantages vis-a-vis other parametric instruments but they do not provide reinstatement of coverage in the case of a pay-out.** CAT bonds are fully collateralized risk transfer mechanisms, free of counterparty credit risk, that provide ex-ante financing without increasing debt and allow for leverage of premium expenditure. A CAT bond will provide important diversification benefits for Jamaica as it triggers in a different way than the CCRIF SPC or the IADB contingent credit line. The lack of mechanisms for reinstatement of coverage in the case of claims is a draw-back of CAT bond instruments, albeit the same applies to contingent lines as they cannot be automatically replenished in the case of disasters. Designing a structure with such a feature would add an additional complexity to the transaction, with Jamaica purchasing reinstatement cover directly from the reinsurance market (with Jamaica managing the procurement of this) or the reinstatement cover being intermediated by the WB (with the WB managing the credit exposure of market counterparties). In other recent WB CAT bond transactions (Pacific Alliance in 2018 and Mexico in 2020), reinstatement cover was considered but was ultimately not purchased due to the additional costs, timing, and complexity.

C. Relevance to Higher Level Objectives

14. **The proposed Project contributes to the World Bank Group's twin goals of ending extreme poverty and boosting shared prosperity by providing fiscal resources in the event of a tropical cyclone which disproportionately affect the poor and vulnerable.** In the aftermath of a disaster, developing countries experience fiscal and liquidity pressures, which affect their ability to provide quality and adequate coverage of public services that the poor and vulnerable need. With relatively quick payouts from the CAT bond, the GOJ will be able to minimize disruptions to the budget and the provision of critical public services, thereby strengthening emergency relief and recovery measures and preventing vulnerable populations from falling deeper into poverty. Ex-ante financial protection through instruments like the CAT bond will help to safeguard the development gains achieved by the GOJ over the recent years.
15. **The Project is well aligned with the objectives of the World Bank Group's Jamaica Country Partnership Strategy (CPS) for FY14-FY17¹⁵** (Report No. 85158-JM) as revised by the CPS Performance and Learning Review (PLR) ((Report No. 112663-JM) both of which are anchored to Jamaica's long-term national

¹⁴ See Technical, Economic and Financial Analysis section for details.

¹⁵ The CPS was extended to 2019 in May 2017. A Systematic Country Diagnostic is currently under preparation.



development plan, “Vision 2030 Jamaica.” The Project objectives are specifically aligned with Pillar III of the CPS on Social and Climate Resilience, focused on expanding the range of financial instruments available to the GOJ to respond to climate change events and natural disasters, thereby building greater financial, social, fiscal, economic growth, and physical resilience. The Project will also help to achieve sustainable economic growth and contribute to Jamaica’s efforts to achieve greener, more resilient, and inclusive development.¹⁶

16. **The Project supports the maximizing of finance for the development agenda of the WB.**¹⁷ This Project will help to further the progress already made by Jamaica on the fiscal front, by securing additional financial coverage from the international markets through crowding in the insurance and capital markets funding to bridge the financial protection gap. This will also support long-term development and open new markets and opportunities for innovative private solutions, for example the insurance-linked securities (ILS) market. This Project will provide an opportunity to test the appetite of the market to subscribe to a CAT bond for the benefit of Jamaica, noting that this would be the first CAT bond for any Caribbean state.¹⁸ The Project is expected to mobilize US\$175-215 million of risk capital from the market.¹⁹ Also, the Project will crowd in private sector expertise including catastrophe risk modelling firms, data collection agencies, investment banks, legal teams, insurance managers, asset managers, reinsurers, reinsurance brokers to help deliver an innovative risk transfer solution.
17. **The Project is consistent with the WB’s strategy to promote innovative products to spread and manage risks.** The background document “New World, New World Bank Group: Post Crisis Directions” prepared for the Development Committee Meeting of April 25, 2010, identifies managing and preparing for crisis as one of the five World Bank Group (WBG) strategic priorities, and states that the WBG’s future work under this strategic priority will focus on “designing innovative finance and insurance products to spread and manage risk” (p.26, para. 61). It is also aligned with the WBG Climate Change Action Plan and its priority to “Leave No One Behind”, particularly paragraph 110 which states that the WBG will “scale up sovereign disaster risk insurance.” The “Forward Look – A Vision for the World Bank Group in 2030” prepared by the WBG for the Development Committee Meeting of October 8, 2016, also focuses on the Global Crisis Response Platform which highlights the importance of regional risk pools and accessing insurance markets. Specifically, the note states that the WBG will provide analytical support to help countries develop “layers of risk financing, from transparent budget contingency planning to the establishment of dedicated reserves for disaster risk management, contingent lines of credit, and insurance-based mechanisms linked to the capital and (re)insurance markets.”
18. **The Project is aligned with the WBG’s Latin American and Caribbean (LCR) Regional Strategy.** The LCR

¹⁶ <http://documents1.worldbank.org/curated/en/136631594937150795/pdf/World-Bank-Group-COVID-19-Crisis-Response-Approach-Paper-Saving-Lives-Scaling-up-Impact-and-Getting-Back-on-Track.pdf>.

¹⁷ Under the current Multilateral Development Bank Reference Guide on Private Investment Mobilization (June 2018), this Project may not be reported as a Private Capital Mobilization Project as it is entirely financed by Trust Fund and there is no WB own financing.

¹⁸ Going forward, a regional CAT bond for Caribbean countries could be issued substantially escalating impact.

¹⁹ The CAT bond coverage amount is indicative and could be higher or lower depending on market pricing when the transaction is executed.



2019 strategy emphasizes the risks and vulnerabilities of the Caribbean region to natural disasters and climate change and the need for greater focus on building 360-degree resilience. The strategy includes a pillar on building resilience that includes managing climate and social risks and managing fiscal and financial risks to which this Project is directly related. Building resilience will also be prominently featured in the new LCR strategy (currently under preparation) to support the implementation of the LCR Climate Action Plan 2021-2025.

2. PROJECT DESCRIPTION

A. Project Development Objective

PDO Statement

To expand Jamaica's financial protection against losses arising from severe tropical cyclones-wind.

PDO Level Indicators

The achievement of the PDO will be measured by the following indicator: Increased financial coverage against natural disasters (US\$130 million).

B. Project Components

19. **This Project has one component that is to develop and implement risk transfer transactions against tropical cyclones-wind effects, including through the financing of transaction costs and premiums.** The Project will pay the risk premium and transaction costs of the CAT bond, a risk transfer instrument issued by the WB through its Treasury for the benefit of Jamaica. As the Project scope is limited to CAT bond design and placement, it does not address the use of CAT bond payouts. Jamaica would enter into an agreement with the WB to pass the catastrophe risk of Jamaica (the "Risk Transfer Agreement", RTA). Contemporaneously, the WB would intermediate the transfer of Jamaica's catastrophe risk to capital market investors by issuing a CAT bond.
20. **In the event of a payout, there may be substantial unused financing that the GOJ could use within the scope of the Project.** When a payout is made from the CAT bond, the coverage from that point forward is reduced by the payout amount, and the risk premiums are reduced in proportion. Any unused financing could be used by Jamaica to replace the coverage by entering into a new risk transfer transaction (insurance or CAT bond), although this may require additional financing to reach the necessary scale. The unused financing may also be cancelled.
21. **Project funds will be disbursed in a payment at the request of the Recipient as specified in the risk transfer legal documents (e.g., Engagement Letter and/or Risk Transfer Agreement) to be signed with WB Treasury (TRE), who will intermediate the risk transfer transaction.** To minimize the operational risk, a single withdrawal request will be made to transfer the funds for the payments on the CAT bond. The



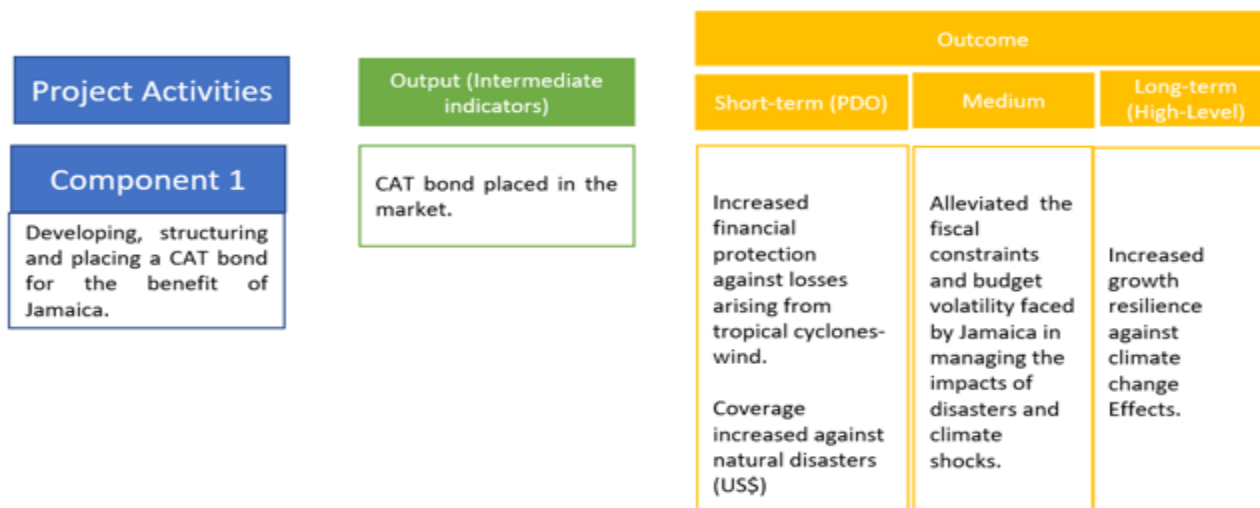
Engagement letter will provide authorization to TRE to begin preparation of the risk transfer transaction and request payment of Project funds to cover risk premium and transaction costs of the CAT bond.

C. Project Beneficiaries

22. **The direct beneficiary of the proposed Project will be the GOJ.** The Project provides financial protection to the GOJ against losses arising from tropical cyclones-wind.
23. **The indirect beneficiaries of the Project will be the population of Jamaica** and, in particular, the poor and vulnerable who are disproportionally affected by disasters. They will ultimately benefit, in the case of an insured event, from the rapid response of the government supported by the insurance payout.
24. **Other indirect beneficiaries will include other Caribbean islands and small states that are interested in exploring capital market instruments to manage their disaster risk.** Small states will benefit from the experience of Jamaica in testing the appetite of the international capital market to supply an innovative capital market transaction as an alternative to index-linked securities. As part of the broader Caribbean program financed by the Disaster Protection Program, Grenada, St. Lucia, and Dominica will also benefit from the data collection and catastrophe risk modelling efforts undertaken in parallel to the work for Jamaica. From this analytical work a regional catastrophe Bond will be explored and if it proves to be cost-effective additional financial support from the Global Risk Financing Facility can be sought to help finance its implementation and premia.

D. Results Chain

25. **The long-term outcome of the operation is to increase growth resilience against climate change effects.** The Project finances the development, structuring, and placement of a CAT bond for the benefit of Jamaica. Assuming investors' interests in investing in the CAT bond, the CAT bond will be placed in the market which will provide increased financial protection against losses arising from TC-W and increase the coverage against those disasters. As the CAT bond provides coverage for up to three years, over the medium term it will alleviate fiscal constraints and budget volatility arising from natural disasters by providing financial resources to manage emergency expenditures related to those events. Enhanced fiscal resilience is expected to translate into increased growth resilience as the availability of fiscal resources to deal with disasters is expected to reduced consumption and production losses associated to natural disaster disruptions.



E. Rationale for Bank Involvement and Role of Partners

26. The WB has supported implementation of several prerequisite activities for this Project funded by the United Kingdom Department Foreign, Commonwealth and Development Office (FCDO) through the “Financial Solutions for Climate and Natural Disaster Risks Program in the Caribbean (P168156)”. The WB’s value proposition includes the provision of specialized technical expertise to support countries in the development and implementation of risk financing solutions and support to countries in building institutional capacity and strengthen government ownership of disaster response. The WB has been providing the GOJ with advisory services independent of any risk carrier, to help the GOJ design a product that enhances their comprehensive DRF strategy and increases their financial resilience. The ongoing TA has supported the GOJ in making informed decisions and is assisting in building capacity in the Ministry of Finance and the Public Service (MOFPS) to manage the fiscal consequences of natural disasters. Specifically, this activity has supported the design of a CAT bond transaction to provide the best coverage and value for money for Jamaica. The activities undertaken by the Project include:

- (i) Commercial CAT modelling of major hazards (earthquake and TC-W), including a survey of public buildings to improve the exposure database conducted by a firm highly regarded by the international capital and reinsurance markets.
- (ii) Recommendations for a comprehensive set of financial instruments and identification of precedence of use.
- (iii) Estimation of the financial gap under different severity scenarios and cost-benefit analysis of a combination of different financing instruments.
- (iv) Recommendations of parametric CAT bond prototypes for earthquakes and tropical cyclone-wind to provide appropriate coverage against low frequency, high impact disasters.
- (v) Trigger improvement and basis risk reduction of the CAT bond.
- (vi) Support for the design, launch, and operationalization of the NNDRF. The TA includes enhancements to the institutional setup, governance arrangements and integration with the fiscal framework.



27. **The WB has successfully secured donor support to cover the CAT bond risk premium and associated transaction costs.** In 2019, the WB began exploring the use of GRiF funds (a multi-donor Trust Fund administered by the WB) to pay the risk premiums and transaction costs (US\$16.365 million) of a CAT bond transaction providing insurance coverage to Jamaica for two or three years. GRiF formally approved the use of resources for this operation in February 2020. The USAID will also contribute US\$5 million to the payment of the risk premiums through a parallel project. For the purpose of this document, all the analytics related to this risk transfer transaction intermediated via CAT bond consider the risk premium and transaction costs payable through this Project and the USAID grant, albeit the PDO indicator considers only what can be achieved with the WB Project funds.
28. **Jamaica has officially requested that the WB intermediate this risk transfer transaction via issuance of a WB CAT bond to benefit from the WB expertise, market brand, and efficient transaction structure.** The GOJ has requested the support of the WB in structuring, marketing, and intermediating the risk transfer transaction. Jamaica does not have experience in executing such transactions. A transaction without the assistance of the WB would involve complexity, time, and cost to establish a Special Purpose Vehicle (SPV) to hold the proceeds of the bonds. The WB has a long history of bringing successful transactions to the market and on account of the WB's expertise and branding, investors are eager to support WB issued CAT bonds.²⁰ This investor eagerness results in WB CAT bonds pricing below the market average. CAT bonds issued by the WB have to date provided clients with US\$3.4 billion of insurance coverage against disasters. Figure 1 illustrates the lower issuance price of WB CAT bonds compared with other outstanding CAT bonds as of August 31, 2020. WB CAT bonds benefit not only from being issued by a known and trusted issuer, but also provide diversifying risks to CAT bond investors. Another means to evaluate WB CAT bonds is to compare the price achieved relative to the initial price guidance provided by dealers during the marketing phase.²¹ An analysis of this data shows that WB CAT bonds price better than dealer guidance and the final price compared to the dealer guidance has reduced more for WB CAT bonds than select market comparators.²² For the US\$2.8 billion of WB CAT bonds issued since 2017, the final pricing came in US\$17.6 million (per annum) lower than the mid-point of the dealer price guidance. In other words, the final pricing of WB CAT bonds has averaged 0.57 percent (unweighted) below the mid-point of the price guidance provided by the dealers. In comparison, final pricing of the selected comparators was on average 0.01 percent below the mid-point of the price guidance provided by the dealers.

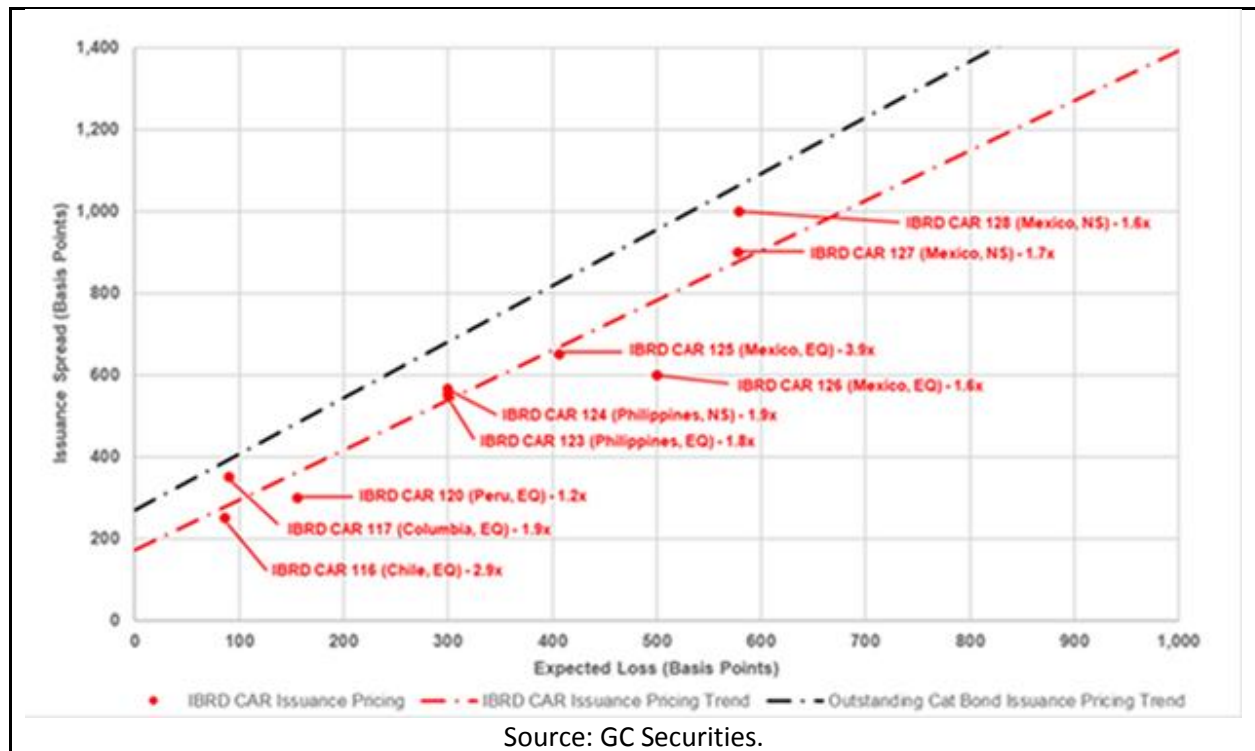
²⁰ No SPV is required for World Bank issued CAT bonds due to the World Bank's issuance of the CAT bond off its regular debt issuance platform.

²¹ Dealers include insurance or reinsurance brokers capital markets units and investment banks.

²² Comparator Bonds analyzed were selected based on the issuance date and perils of World Bank CAT bonds.



Figure 1: Issuance price of IBRD and other outstanding CAT bonds as of Aug 31, 2020



29. **A waiver was obtained to allow the WB to have an expanded implementation role under the Project and intermediate the risk transaction.** This expanded WB role does not involve typical Project implementation. Intermediation constitutes a financial transaction executed at the request of the Recipient and is similar to conversion options provided by the WB under IBRD flexible loans. The WB will not be implementing activities on Jamaican territory or purchasing goods to be delivered to Jamaica. The WB is authorized to provide these risk intermediation services under its existing policies applicable to disaster risk management products offered to clients as approved by the Board of Executive Directors. To mitigate any potential conflict of interest or perceived conflict of interest that would have arisen from using Project funds to pay the IBRD intermediation fee, Jamaica has allocated budgetary resources in this fiscal year to cover this expense. Hence, no GRiF funds would be directed to cover WB costs.
30. **The proposed operation does not use the country's IBRD lending envelope while helping the WB to optimize the use of its financing.** The risk transfer transaction intermediated via CAT bond is structured to avoid credit exposure to Jamaica. The Jamaican catastrophe risk is fully passed to investors through the issuance of a WB CAT bond. Thus, there is no use of the WB envelope for Jamaica or counterpart credit exposure. Since the proposed Project provides fiscal resources to mitigate the impact of natural disasters, it reduces future borrowing needs for this purpose, including from the WB.
31. **This Project builds on existing in-country operations and dialogue in various country engagements with**



units across the WB. The GOJ has embarked upon a comprehensive strategy toward developing 360-degree resilience to natural disasters and climate shocks. The reforms cover fiscal, financial, physical, and social resilience (See Annex 4). Some of these are being supported by the WB especially through policy lending and technical assistance. The reforms are in varying stages of completion and are being supported by the following: (i) the WB series of two Development Policy Financing (DPFs) to build long-term fiscal and financial resilience (DPF1 for US\$70 million already approved) and (ii) a COVID-19 Response and Recovery DPF in the amount of US\$150 million (approved in March 2021).

F. Lessons Learned and Reflected in the Project Design

32. **The experience of developing and intermediating risk transfer transactions has offered several important lessons that have been used to inform the Project design for Jamaica.** The WB has intermediated risk transfer transactions (insurance and CAT bond) for several countries, including Mexico, Philippines, Colombia, Chile, and Peru. Lessons from these interventions include the following:
33. **CAT bonds need to be integrated into the overall DRF strategy and to act as a complement to the existing suite of financial instruments.** Risk transfer instruments such as insurance and CAT bonds cannot solve all the challenges associated with a country's vulnerability to climate related risk. Recognizing this, the Project has been designed using a risk layering strategy with a combination of instruments to cover tropical cyclone events of different severity. The CAT bond forms a critical part of the GOJ's risk layering strategy and complements the other instruments such as the existing Contingencies Fund for Natural Disasters, the contingent line of credit from the IADB, and CCRIF insurance policies to strengthen the financial resilience of the GOJ.
34. **Capacity building and technical support are important for successful implementation of a market-based risk transfer instrument.** Capacity building is needed to increase understanding of the financial structure of the product and to ensure that the instrument fully meets their needs. In addition, it helps to set clear expectations on when a payout would be triggered and, more importantly, when it would not. This Project has benefitted from the FCDO-supported technical assistance project under which an analysis of the existing financial instruments held by Jamaica was conducted to demonstrate where the CAT bond would be most cost effective. This Project has also built internal capacity at the MOFPS, including the placement of a quantitative resident economist, to support the ongoing assessment of the GOJ's contingent liabilities caused by disaster and climate risks and the implementation of risk transfer instruments such as CAT bonds.
35. **The trigger used to determine payouts should be transparent, easily understood, and verifiable.** The trigger design can leverage structures developed by others and refined over time. For example, Mexico has been refining their parametric trigger structure since their first CAT bond issuance in 2006 (from CAT-in-a-box to CAT-in-a-Grid to a refined CAT-in-a-Grid). This experience will be leveraged to develop a trigger that best fits the needs of the GOJ. The process for optimizing the trigger structure has also evolved over time and this Project will utilize latest developments in the area of optimization.



3. IMPLEMENTATION ARRANGEMENTS

A. Institutional and Implementation Arrangements

36. **The MOFPS will be the recipient and implementing entity for this operation.** The proceeds of the Grant under this Project will finance the payments of risk premium and transaction costs related to the design, structuring, and placement of the CAT bond for supporting the GOJ to expand its financial protection against losses arising from tropical cyclones. Prior to Project approval, the MOFPS will be responsible for requesting the WB's intermediation (i.e., making an independent decision on engaging the WB), providing input to the WB on the design and objectives of the risk transfer transaction, and signing the legal documentation for the risk transfer transaction (e.g., RTA). During Project implementation, the MOFPS will be responsible for the monitoring and evaluation of arrangements that facilitate proper reporting, and for submitting withdrawal requests to the WB. The proceeds of the Grant will be disbursed from the Grant Account to TRE according to the terms and conditions of the Grant Agreement.
37. **The WB, through its Treasury, will intermediate the risk transfer transaction.** The WB will enter into the RTA (insurance or derivative type contract) with the GOJ, providing Jamaica with insurance cover against pre-defined catastrophe events. The WB will issue a CAT bond to transfer the Jamaica catastrophe risk to capital market investors. The terms of the RTA and the CAT bond are aligned, so if a payout to Jamaica is triggered, the CAT bond principal will be reduced by the same amount. The Grant proceeds will be used to pay the risk premium of the CAT bond and the transaction costs under the terms of the RTA and related documentation between the WB and Jamaica. TRE will engage external service providers, such as external legal counsel, lead arrangers (managers, structuring agents, bookrunner agents), event calculation agent, and listing agent to issue the CAT bond and to prepare the legal documentation for the risk transfer transaction.
38. **Intermediation by TRE occurs in a highly regulated environment.** TRE is using lead dealers, makes the prices public, and charges the standard IBRD intermediation fee that applies irrespective of whether the premium is funded by a WB Project or directly by a country. Events that trigger the payout are known well in advance and verified by an independent third-party (event calculation agent), selected through a competitive procurement process, who consolidates public information received from independent agencies. There is therefore no scope for interpretation or negotiation.
39. **It is proposed to implement the Project over a three-year period.** The CAT bond will be placed by end July 2021 and mature Dec. 21st, 2023, providing coverage for three hurricane seasons. Project closing at end June 2024 will allow sufficient time to calculate payouts, premiums and risks in case of a late event in December 2023.²³

²³ In the remote event of a reporting agency failure by the National Hurricane Centre, then the bond could be extended by up to 4 months, and in this remote scenario a payout would come later than that set out in the payment timeline. This remote scenario is allowed for in the bond documentation and the RTA. During this extension period, extension premiums would be due to bondholders and some small transaction costs will be incurred (Intralinks fees).



40. **Premium Financing.** The GRiF has approved a grant to fund risk premiums up to US\$14.85 million for a CAT bond plus transaction costs estimated at US\$1.515 million, for a total grant of US\$16.365 million. USAID will provide additional US\$5 million grant funding towards risk premiums of the CAT bond, which would allow for an increase in the coverage or term of the CAT bond. This would bring the total risk premium financing to US\$19.85 million. The USAID will directly pay the premium contribution to TRE under a parallel project.

B. Results Monitoring and Evaluation Arrangements

41. **The WB will complete periodic Implementation Status and Results Reports as well as an Implementation Completion and Results Report within six months following the end of the Project.** The Bank will collect all necessary data and report on a semi-annual basis on the use of grant proceeds and Project performance, including information on intermediate Project results and progress toward higher-level outcomes.
42. **The Bank will provide strong technical and operational oversight and supervision of the Project activities.** In addition, given that the transaction is executed in a highly regulated capital market (agents involved in the transaction must comply with international capital markets regulations and controls), there will be an enhancement in oversight of the transaction.

C. Sustainability

43. **The sustainability of this Project and other DRF-related efforts is being demonstrated by the GOJ in three main ways: financial, human resource and institutional sustainability.**
- **Financial sustainability:** The CAT bond instrument is part of the GOJ's efforts to build a comprehensive risk financing strategy. Should this instrument demonstrate positive results, it is intended that the GOJ will finance future premiums through its own budget after the first three years of donor funding. The GOJ has already made strides in drawing on its own budget for financing costs associated with other financial instruments and continue to create the fiscal space to be able to fund on its own its DRF instruments. The latter could include potential future CAT bond risk transfers.
 - **Building of in-house technical capability:** The government is making strides in developing its own capacity in disaster risk management and DRF so it can manage programs on this topic and make it central to government financial planning and development processes. Also, the FCDO-funded program is currently supporting staffing as well as capacity building delivered by a policy expert and an international quantitative economist (resident advisor) to support the MOFPS in the preparation, evaluation, and implementation of future financial instruments. The resident advisor will also provide training to the local staff, ensuring sustained and long-term capacity building and knowledge transfer for GOJ.
 - **Strengthening of institutions and systems to embed risk financing within broader risk management efforts:** To mitigate the financial impact of natural disasters, GOJ has embarked on a comprehensive



approach to disaster risk management and preparedness, which includes strong policy reforms, development of robust instruments, and institutional and capacity building. A national policy on disaster risk financing was approved by the Cabinet in November 2018 and is expected to be submitted to Parliament for approval by June 2021. Specifically, to increase transparency and accountability, as well as effective flow of funds, reforms are ongoing for (i) post-disaster budget execution guidelines to expedite funding for disaster relief and recovery; (ii) a new Procurement Act to ensure timely procurement in the event of a disaster, including expedited procurement for emergency situations; (iii) expansion of the main conditional cash transfer program and advancement of the Humanitarian Assistance Plan and Strategy (HAPAS) to provide timely support to poor and disaster-affected households; and (iv) strengthening of physical resilience through several reforms, including the issuance of new building codes.

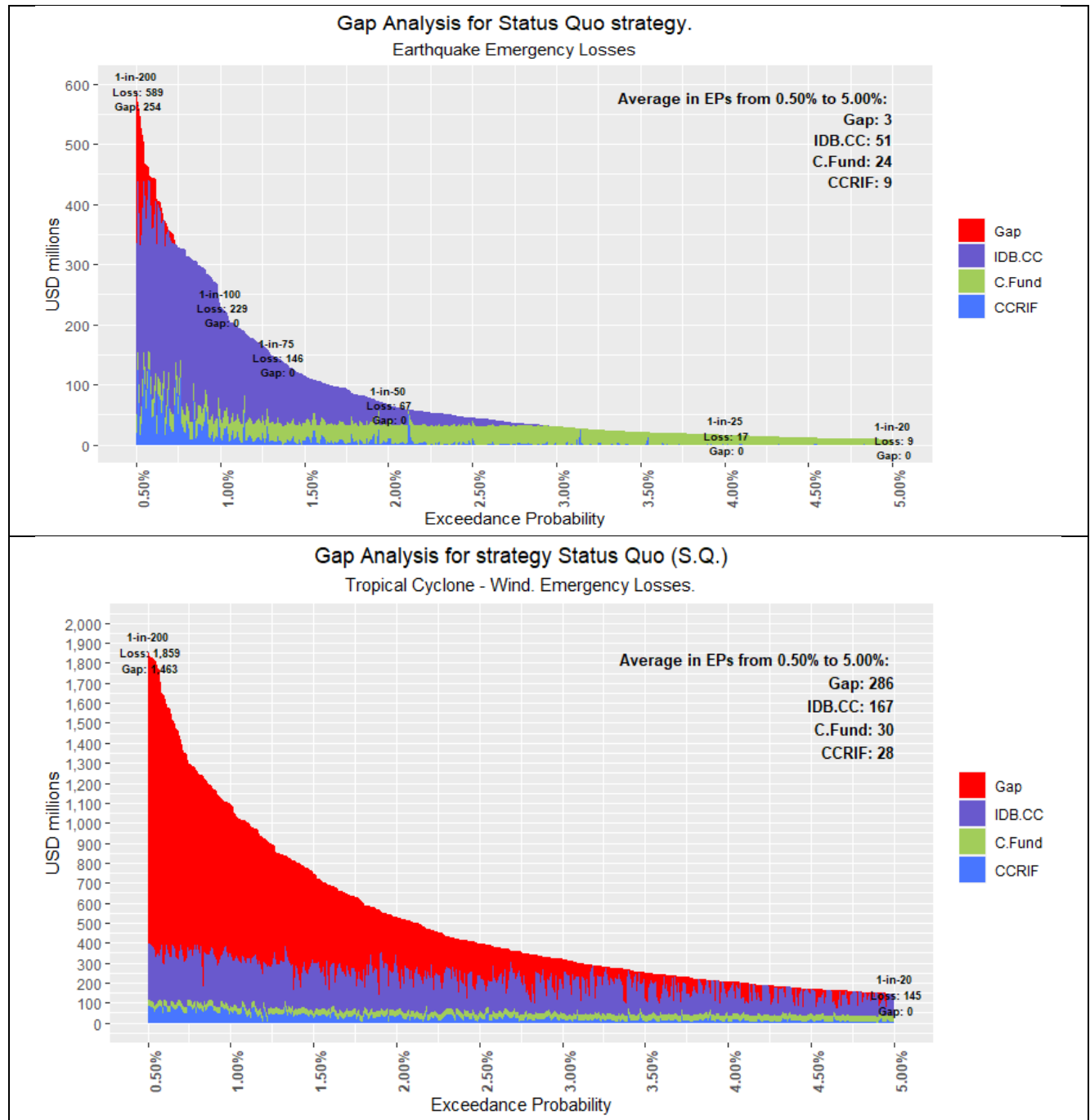
4. PROJECT APPRAISAL SUMMARY

A. Technical, Economic and Financial Analysis

- 44. The study of the CAT bond as an additional risk transfer instrument is focused on TC-W emergency losses.** The financial gap for earthquake is smaller than for tropical cyclone; for the former, Jamaica is buying almost 100 percent of the total coverage available but for the latter, Jamaica is buying a small fraction of the total coverage available. The emergency losses related to earthquake are very well covered almost across the complete loss distribution, except for an annual loss exceedance probability of 0.75 percent, i.e., the residual risk for which does not makes sense to buy ex-ante risk transfer instruments. For TC-W, the gap is also massive for the range of annual exceedance probabilities lower than 1 percent. For the range of annual exceedance probabilities between 5 percent and 1 percent (comprising low frequency and high severity events), there is an important sized gap that may be managed by risk transfer instruments. Finally, for the range of annual exceedance probabilities higher to 5 percent (which includes high and middle frequency and low and middle severity events) the gap is almost inexistent or zero.



Figure 2: Gap Analysis (Emergency losses- EQ and TC-W) for Status Quo Strategy

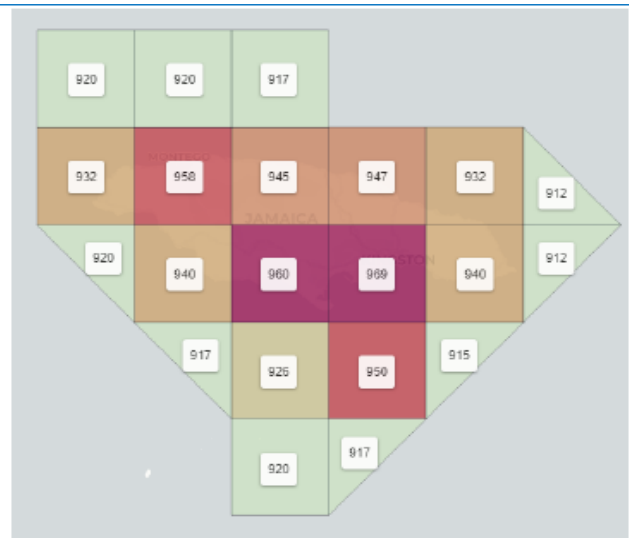




45. **Seven CAT bond prototypes have been developed, analyzed, and the GOJ has decided on a preferred structure.**

The Project benefits from TA related to the cat modelling, generation of CAT-Bond prototypes and advice on best options. CAT bond prototypes for Jamaica were prepared by AIR in 2020 and improved in 2021. The TC-W coverage will trigger on the central pressure of the tropical cyclone as reported by the U.S. National Hurricane Centre. The proposed solution covers the geographic area of Jamaica using a grid with 21 cells and with the improved grid proposed the number of historical events that trigger a payout increased to 8 from 6 on the initial prototype. If the track of a tropical cyclone crosses through one or more boxes, then a payout might be triggered according to the minimum central pressure of the hurricane in the box. Each box has, at least, two trigger thresholds which trigger different percentage payouts of the CAT-Bond based on a stepwise linear function.²⁴ For example, in one of the prototypes, the cell that includes Kingston is parameterized as follows:

- Payout from 30 percent to 70 percent if central pressure is between 969mb and 900mb,
- Payout from 70 percent to 100 percent payout if central pressure is less than 900mb,



46. **The analysis assumes the GRiF grant may be allocated into three instruments as follows:**

- **CAT bond for tropical cyclone (wind)** with an annual premium of US\$7.94 million for 2.5 years.²⁵ The CAT bond was evaluated for seven different prototypes and for pricing purposes, the multiples²⁶ were assumed at the mid-value of the indicative range provided by Lead Managers. Initially, a harder market condition multiple was analyzed, but the value-for-money are, as expected and demonstrated in a previous analysis, worst. By June 2021, the ILS market is recovered and offering reasonable pricing.
- **Expand the Contingencies Fund** by a lump-sum of US\$19.85 million, and
- **Expand the TC-W CCRIF SPC's** coverage with an annual premium of US\$6.62 million for three years.

Initially, nine competing financing strategies²⁷ were evaluated performing a *funding gap* analysis and a *cost-benefit analysis*, but as the new pricing information arrived, some of the strategies were eliminated and the following eight strategies were evaluated:

²⁴ The stepwise function provides for a linear interpolation between the ranges.

²⁵ During the evaluation of the prototypes, it was decided that the Cat-bond will have a duration of 2.5 years (coverage termination on December 21st, 2023) as a way to help to increase the Cat-Bond value while protecting 1-extra month after the "official" end of the hurricane season.

²⁶ The multiple is defined as the ratio of premium paid to expected loss.

²⁷ The results of this evaluation are offered in the previous Technical Note, dated April 4th, 2021.



Financing Strategy*	
1.	No DRFI Strategy
2.	Status Quo (S.Q.)
3.	S.Q. and Expand C. Fund
4.	S.Q. and Expand CCRIF
5.	S.Q. and CAT bond (EL 1.20%, AP 1.98%, EP 0.5%)
6.	S.Q. and CAT bond (EL 1.48%, AP 2.42%, EP 0.74%) - Preferred
7.	S.Q. and CAT bond (EL 1.55%, AP 2.55%, EP 0.81%)
8.	S.Q. and CAT bond (EL 3.05%, AP 4.08%, EP 1.74%)

* A financing strategy is defined as a portfolio of different financial instruments with a pre-defined usage precedence.

47. The CAT bond limit, or maximum potential payout, for the preferred prototype (EL = 1.48%) ranges from US\$175 million to US\$215 million (US\$193 million average), while the additional Limit in case of CCRIF SPC's expansion (EL = 2.26%) is about US\$184 million. It is important to note that CCRIF SPC TC insurance coverage is smaller for this underwriting year driving by the reduction of its attachment point and the higher rate on line (3.11% vs. 3.58%) produces a lower leverage.²⁸
48. From the *financing gap analysis*, it was found that, for all strategies, the gaps remain massive for very high severe events as measured by *tail value at risk at 1percent (TVaR_x²⁹)*, and the losses in this range should not be considered as part of ex-ante risk transfer instruments. For the tail's middle range (annual exceedance probabilities between 5 percent and 1 percent (low frequency and high severity events) it is derived that some CAT bond prototypes produce gaps similar to the strategy "4. S.Q. and Expand CCRIF". For the range of probabilities higher than 5 percent (high and middle frequency and low and middle severity events) the gap is almost inexistent or zero with the status-quo strategy.

Strategy	Financing Gap		
	Average	Cond. Ave. 1% to 5%	TVaR 1%
1. No DRFI Strategy	40.82	400	2,097
2. Status Quo (S.Q.)	25.08	191	1,727
3. S.Q. & Expand C. Fund	24.06	174	1,707
4. S.Q. & Expand CCRIF	21.47	145	1,558
5. S.Q. & CAT bond (EL 1.20%, AP 1.98%, EP 0.5%)	22.67	165	1,588
6. S.Q. & CAT bond (EL 1.48%, AP 2.42%, EP 0.74%) -Preferred	22.59	162	1,592
7. S.Q. & CAT bond (EL 1.55%, AP 2.55%, EP 0.81%)	22.52	161	1,591
8. S.Q. & CAT bond (EL 3.05%, AP 4.08%, EP 1.74%)	22.35	150	1,622

49. The *Cost-Benefit Analysis* (optimization of Net Present Value of Opportunity Cost) indicates that:

²⁸ Rate on line is the percentage derived by dividing reinsurance premium by reinsurance limit.

²⁹ TVaR is the conditional expected losses with exceedance probabilities lower than "x".



- a. The Strategy “4. S.Q. and expand CCRIF” optimizes NPV when looking up to the most severe events as evaluated with the TVaR at 1 percent. Given the design of the CCRIF TC and the preferred Cat-bond prototype coverages, full payouts would only be triggered in the most severe events (exhaustion probability is targeted to around 1-in-150 years).
- b. For the high-severity and low-frequency layer (exceedance probabilities between 1 percent and 5 percent), some strategies that include CAT bond optimize NPV. The strategies that perform better include prototypes that help to reduce gaps for emergency losses with exceedance probability between 5 percent and 1 percent.
- c. For less severe and high frequency layer (exceedance probabilities greater than 5percent), the status quo is the best strategy, because the incurred costs of the alternative strategies are not compensated by cash inflows from the new/expanded instruments. In this case, TC-W emergency losses would be properly financed with the current instruments in place.

Strategy	Net Present Value		
	Average	Cond. Ave. 1% to 5%	TVaR 1%
1. No DRFI Strategy	-5.24	-51	-269
2. Status Quo (S.Q.)	-2.73	27	-96
3. S.Q. & Expand C. Fund	-3.07	28	-94
4. S.Q. & Expand CCRIF	-4.81	76	88
5. S.Q. & CAT bond (EL 1.20%, AP 1.98%, EP 0.5%)	-7.63	50	53
6. S.Q. & CAT bond (EL 1.48%, AP 2.42%, EP 0.74%) -Preferred	-7.53	54	49
7. S.Q. & CAT bond (EL 1.55%, AP 2.55%, EP 0.81%)	-7.47	57	50
8. S.Q. & CAT bond (EL 3.05%, AP 4.08%, EP 1.74%)	-6.84	71	14

50. Considering the results of both the Gap and Cost-Benefit analysis, the selection of the best strategy depends on the risk layer targeted by the GOJ as there is a tradeoff between the probability of receiving a payout and its size. The decision made by the GOJ (EL=1.48%, AP=2.42%, EP=0.74%) is a prototype which is between two “extreme” prototypes: one offering frequent but smaller payouts (EL=3.05%, AP=4.08%, EP=1.74%) and the other one offering rare but higher payouts (EL=1.20%, AP=1.98%, EP=0.50%)
51. The chosen prototype (EL=1.48%, AP=2.42%, EP=0.74%) provides value-for-money in comparison with the status-quo strategy. It is important to bear in mind that the expansion of the CCRIF SPC TC insurance is theoretical as there is no guarantee that the company is able and/or willing to provide such sum



insured, in fact, Jamaica reduced its attachment point for the underwriting year 2021-2022 signaling that they are taking advantage of CCRIF lower price for more frequent events. On the other hand, it should be noted that the CAT bonds will diversify the basis risk of the CCRIF insurance policies and the IADB contingent credit, an important benefit not considered in the NPV calculations.

52. **These results are highly sensitive to the assumed multiple for the CAT bond prototypes and must be revised once market pricing has been received during the bidding process to inform the best structure of the transaction immediately prior to placement. It is also recommended to keep fine-tuning the potential CAT bond structure.**
53. **Finally, it is important to stress the fact that the analysis and conclusions are only quantitative based thus it does not acknowledge relevant qualitative aspects.** For example, a basis risk potential reduction of the financing strategy because the pure parametric trigger of the CAT bond will complement the modelled-loss trigger of the CCRIF insurance policy; that is, by using a different trigger mechanism in the CAT bond, some element of basis risk may be reduced.

B. Fiduciary

Financial Management

54. **Overall financial management risk is expected to be low.** There will be no transfer of Project funds to the GOJ, as the GRiF-funded risk premium and transaction costs for the CAT bonds will be directly disbursed and paid by TRE to the CAT bond investors and service providers. TRE will report actual expenditure/payments to the implementing entity so that documentation of expenditure can be provided by the implementing entity within 30 days of each calendar quarter. A format for the Statement of Expenditures has been agreed and enclosed in the Disbursement and Financial Information Letter (DFIL) of the Project as attachment 2. Each Project report will include information on receipts and expenditures documented in client connection. Therefore, no interim financial reports are required. Unused proceeds, if any, at the close of the Project will be returned to the GRiF. Considering that TRE is under WB's internal control framework, there is adequate assurance that the funds will be used for the intended purposes. Elimination of audit requirements has been approved by the Financial Management Operations Review Committee (FMORC). Therefore, no annual audited financial statements are required.

Disbursement Arrangements

55. **The proceeds of the Grant will be disbursed from the WB Grant Account to TRE** according to the terms and conditions of the risk transfer legal documentation (i.e., the Engagement Letter and/or the RTA). The payment method to be used will be "advances" into a designated account managed by TRE. The eligible expenditures will be payments made for risk premium and transaction costs, as set out in the Grant Agreement and the risk transfer legal documentation. Risk premiums and transaction costs related to the design, structuring and placement of the CAT bond will be reported by TRE to the implementing agency who would then report to the WB to document eligible expenditures. The percentage of expenditures to be financed, inclusive of taxes, is 100 percent. The documentation of the eligible expenditures, using



Statements of Expenditures, and the detailed disbursement arrangements will be described in the DFIL of the Project.

Procurement

56. **There will be no contracts for goods, works, consulting and non-consulting services subject to WB procurement regulations for IPFs.** The Project will finance risk premium and transaction costs incurred by TRE for the CAT bond only. Any procurement conducted by TRE will be done according to the Bank's corporate procurement policies.

C. Legal Operational Policies

	Triggered?
Projects on International Waterways OP 7.50	No
Projects in Disputed Areas OP 7.60	No

D. Environmental and Social

57. **The WB has assessed the environmental and social risk of the Project as “Low”.** This Project has one component that is to develop and implement risk transfer transactions against tropical cyclones-wind effects, including through the financing of transaction costs and premiums. The scope of the Project is limited to payment of the risk premium and transaction costs of the CAT bond and, accordingly, the Project will not have any adverse impacts on the population and the environment. Therefore, the Environmental and Social Framework will apply only to this risk premium and related transaction costs, not to the payout. Given the limited focus of the Project on the premium, the Project's environmental and social risks is assessed as Low Risk under the WB's Environmental and Social Policy.
58. **Considering the scope of the Project, labor risks are minimal.** The WB will intermediate the transaction through TRE. A limited number of civil servants from the MOFPS are expected to work on the Project. The ESS2 Labor and Working Conditions will apply to the civil servants of the MOFPS assigned to work on the Project with respect to its launch and placement on the market. Where civil servants are working in connection to the Project, they will remain subject to the terms and conditions of their existing public sector employment agreement. Only the provision of ESS2 that relates to Occupational Health and Safety (OHS) will apply. Given the very limited number of civil servants expected to work on the Project, no separate LMP will be prepared. The Project, through the Environmental and Social Commitment Plan (ESCP), will be required to confirm that the Government of Jamaica National OHS requirements are undertaken in accordance with the requirements established in ESS2 - Labor and Working Conditions and the Labor Laws in Jamaica within 90 days of Project effectiveness.
59. **ESS 10 Stakeholder Engagement and Information Disclosure is relevant.** The launch of the CAT bond and its placement will be managed through TRE, including the communication with potential investors



in the global Bond market. The MOFPS has prepared and disclosed prior to appraisal a Stakeholder Engagement Plan (SEP) that will include the proposed stakeholder engagement activities throughout the Project-life cycle and a feedback mechanism. The MOFPS has an experienced Communication Department led by a Manager of Corporate Relations to oversee the SEP implementation in Jamaica. The stakeholders in Jamaica include the general public, the Office of Disaster Preparedness and Emergency Management, the National Disaster Committee, National Environment and Planning Agency (NEPA), Ministry of Local Government and Rural Development, Ministry of Economic Growth and Job Creation, Ministry of Health and Wellness and its players and the relevant ministries and general public.

60. **Status of Instruments.** The assessment done at concept stage and during Project preparation identified ESS1, ESS2, and ESS10 as relevant and will focus only on the payment of the premium and transaction costs. Given limited scope of the Project and no potential environmental and social risk and impact associated with the Project activities, no further assessment was necessary in accordance with ESS1.

- a) *ESS1*: The Recipient has prepared a draft ESCP, disclosed on the Bank's website on June 8, 2021.
- b) *ESS2*: The MOFPS has in place OHS protocol in regard to COVID-19 and these will be maintained throughout Project implementation. MOFPS staff assigned to the Project will retain their positions and will channel any grievance through the MOFPS Human Resources Department which has a complaint mechanism.
- c) *ESS10*: The Recipient has prepared a draft SEP, disclosed on the Bank's website on June 8, 2021 focused on potential investors and other institutions, including regulators and parties involved in GOJ's fiscal and emergency management. There are two feedback and GRMs (domestic and international). The former is in place to be managed by the MOFPS Corporate Communications and the latter will be in place after a firm selection to place the Bond has commenced and a prospectus is finalized. The SEP will be updated and adopted within 90 days of Project effectiveness.

61. **In a broader context, the WB is supporting the client's capacity vis-à-vis environmental and social risk management in a disaster context.** The Ministry of Finance recently executed the Jamaica First Economic Resilience DPL (P170223 approved Feb 2020), which had explicit sections on the capacity of GOJ in respect of environmental and social capacity and fiscal capacity. While these activities are outside the scope of this Project, they demonstrate how the government is managing environmental and social risks at a policy level.

62. **Jamaica has an established environmental management system in-country.** Under the Natural Resources Conservation Authority (NRCA) Act of 1991, the NRCA was authorized to issue, suspend, or revoke permits and licenses for non-compliance with the stipulated environmental standards and conditions. Sections 9 and 10 of the NRCA Act stipulated that an Environmental Impact Assessment is required for new projects and existing projects undergoing expansion. The NRCA was also responsible for investigating the effect on the environment of any activity that may cause pollution or which involves waste management. The NEPA became operational on April 1, 2001, as the lead government agency with the mandate for environmental protection, natural resource management, land use, and spatial planning in Jamaica and took the responsibility of NCRA.



63. **Given the nature of the Project, few grievances or complaints are anticipated. Rather it is expected that the Grievance Redress Mechanism (GRM) will be utilized as a feedback mechanism to receive queries from the public for information about the instruments and its functioning. The Project will maintain a feedback mechanism to receive any questions or concerns that may arise.** This feedback mechanism is detailed in the SEP and is the existing feedback mechanism that is employed by the MOFPS, which is managed by the Corporate Communications Department. The MOFPS has an in-house social media specialist, who monitors the various platforms and addresses any issues and concerns arising. The social media specialist reports to the Manager, Corporate Communications, who is ultimately responsible for the implementation of SEP and GRM. The Feedback and GRM for international stakeholders will be operational once the prospectus document for the CAT bond is completed and the lead firm selected to issue and manage the Bond will be the contact point for queries and concerns.

5. GRIEVANCE REDRESS SERVICES

64. **Communities and individuals who believe that they are adversely affected by a WB supported Project may submit complaints to existing project-level grievance redress mechanisms or the WB's Grievance Redress Service (GRS).** The GRS ensures that complaints received are promptly reviewed in order to address Project-related concerns. Project affected communities and individuals may submit their complaint to the WB's independent Inspection Panel which determines whether harm occurred, or could occur, as a result of WB non-compliance with its policies and procedures. Complaints may be submitted at any time after concerns have been brought directly to the WB's attention, and WB Management has been given an opportunity to respond. For information on how to submit complaints to the WB's corporate Grievance Redress Service, please visit <http://www.worldbank.org/en/projects-operations/products-and-services/grievance-redress-service>. For information on how to submit complaints to the World Bank Inspection Panel, please visit www.inspectionpanel.org.

6. KEY RISKS

65. **The overall risk to the achievement of the PDO is assessed as Moderate.** The moderate risk rating is driven by global macroeconomic volatility that could hamper placement of CAT bond if ILS markets are affected, and technical design of the Project given the complexity involved in designing CAT bond prototypes and the envisioned tight timeline for CAT bond placement. There is a minimal risk the product does not trigger payouts when expected by Jamaica that can be managed by several factors. The payout will be determined by a relatively simple parametric trigger, discussed and agreed with the GOJ. Moreover, the preceding programmatic advisory services Project not only helped identify the appropriate disaster risk transfer, but also helped the GOJ understand the nature of parametric insurance and triggers. In addition, during the placing, the WB will work with an experienced risk modeling firm, work with experienced CAT bond dealers (selected through a competitive selection



process), work with an experienced external legal counsel in the CAT bond market and leverage the WB's own experience in the CAT bond market. As the Project is limited to payment of the CAT bond premium and transaction costs, and the CAT bond is placed by TRE, which directly receives Grant Resources, safeguards and fiduciary risks are low. Jamaica is firmly committed to the Project.



7. RESULTS FRAMEWORK AND MONITORING

Results Framework

COUNTRY: Jamaica

Jamaica Catastrophe Bond for increased Financial Resilience to Natural Disasters and Climate Shocks

Project Development Objectives(s)

To expand Jamaica's financial protection against losses arising from severe tropical cyclones-wind.

Project Development Objective Indicators

Indicator Name	PBC	Baseline	End Target
Increased financial coverage against natural disasters			
Increased insurance coverage (Amount(USD))		0.00	130,000,000.00

Intermediate Results Indicators by Components

Indicator Name	PBC	Baseline	End Target
Development and implementation of risk transfer transactions against tropical cyclones-wind effects			
CAT bond placed in the market (Yes/No)		No	Yes



Monitoring & Evaluation Plan: PDO Indicators

Indicator Name	Definition/Description	Frequency	Datasource	Methodology for Data Collection	Responsibility for Data Collection
Increased insurance coverage	Increased insurance coverage of approximately US\$ 130 million over an above existing coverage.	Once at inception and then annual thereafter if no insured event occurs	Ministry of Finance and the Public Service of Jamaica	Confirmation from the Ministry of Finance and the Public Service on its suite of financial instruments and level of coverage	World Bank Task Team

Monitoring & Evaluation Plan: Intermediate Results Indicators

Indicator Name	Definition/Description	Frequency	Datasource	Methodology for Data Collection	Responsibility for Data Collection
CAT bond placed in the market					



ANNEX 1: Implementation Arrangements and Support Plan

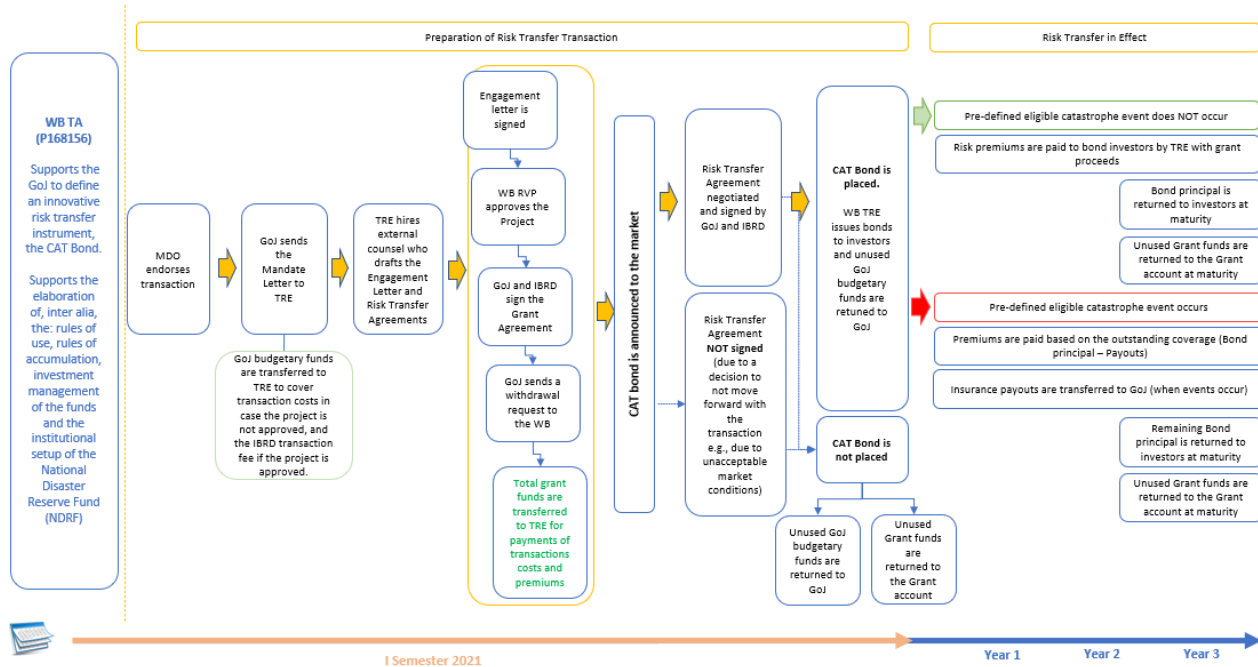
COUNTRY: Jamaica

Jamaica Catastrophe Bond for increased Financial Resilience to Natural Disasters and Climate Shocks

Institutional and Implementation Arrangements

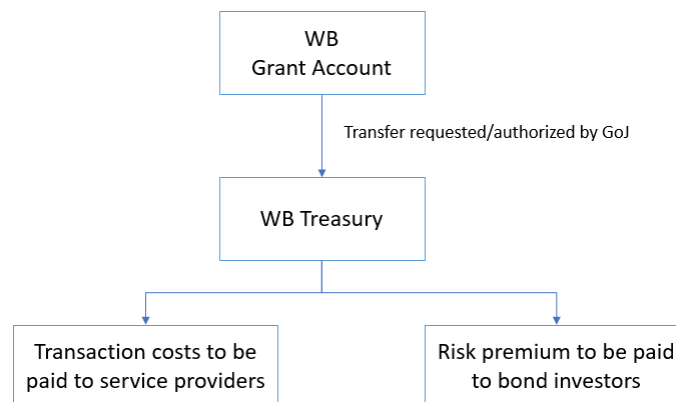
- 1. The Ministry of Finance and the Public Service (MOFPS) will be the recipient and implementing entity for this operation.** The proceeds of the Grant under this Project will finance the payments of risk premium and transaction costs related to the design, structuring and placement of the CAT bond for supporting the GOJ to expand its financial protection against losses arising TC-W. Prior to Project approval, the MOFPS will be responsible for requesting the WB's intermediation (i.e., making an independent decision on engaging the WB), providing input to the WB on the design and objectives of the risk transfer transaction and signing the risk transfer legal documentation. During Project implementation, the MOFPS will be responsible for the monitoring and evaluation of arrangements that facilitate proper reporting and for submitting withdrawal requests to the WB. The proceeds of the Grant will be disbursed from the Grant Account to TRE according to the terms and conditions of the Grant Agreement.
- 2. At the request of the GOJ, the WB, through its Treasury, will intermediate the risk transfer transaction.** The risk transfer transaction process for this operation responds to the GOJ request for advancing promptly in the design, structuring and placement of the CAT bond. In the case of this operation, after the WB endorses the preparation of this transaction, the GOJ sends a Mandate Letter to TRE requesting to start the preparation of the risk transfer transaction and transfers budgetary funds to cover transaction costs in case the Project is not approved, and the IBRD transaction fee, if the Project is approved. TRE hires an external counsel who drafts the legal documentation for the risk transfer transaction, which are then negotiated and signed by the GOJ and IBRD. After the WB approves the Project and the Grant Agreement is fully executed, the GOJ will submit a withdrawal application to the WB requesting the transfer of the total Grant proceeds to TRE to finance the payments of transaction costs related to the design, structuring and placement of the CAT bond and the risk premiums. After the CAT bond design is approved by the GOJ, TRE announces the CAT bond to the market. If the Bond is not placed, the unused GOJ budgetary funds are returned to the government, and the unused Grant funds are returned to the Grant account. If the CAT bond is placed, TRE issues the Bond to investors and the unused GOJ budgetary funds are returned to the government. After the risk transfer is in effect, if the pre-defined eligible catastrophe event does not occur, risk premiums are paid to Bond investors by TRE with Grant proceeds according to the terms and conditions of the standard market documentation for CAT bond issuance. At Bond maturity, the Bond principal is returned to investors and any unused Grant funds are returned to the Grant account. If a pre-defined eligible catastrophe event occurs, insurance payouts are transferred to the GOJ and risk premiums are paid to Bond investors by TRE with Grant proceeds based on the outstanding coverage (Bond principal – Payouts). At Bond maturity, the outstanding Bond principal is returned to investors and the unused Grant funds are returned to the Grant account.

Figure 1: Risk Transfer Transaction Process for the Jamaica CAT bond



3. The flow of funds for this operation will include the following:

Figure 2: Flow of Funds of Grant Proceeds



- The Grant proceeds will be disbursed to the TRE as instructed by the GOJ according to the terms and conditions of the DFIL and consistent with risk transfer legal documentation (e.g., the Engagement Letter and/or the Risk Transfer Agreement) between the GOJ and the WB.
- Risk premiums for the insurance cover will be paid from the proceeds of the Grant, and fully passed to the Bond investors by TRE.



- (c) Transaction costs (excluding the IBRD 10 bps intermediation fee) for the risk transfer transaction will be paid to TRE from the proceeds of the Grant, and fully passed to the external service providers by TRE.
4. **The WB prepares and monitors the Project.** As part of Project preparation, the WB is working with the GOJ Representatives through technical assistance projects building capacity for disaster risk management, modelling catastrophe risks and designing the CAT bond prototypes. The WB will monitor Project performance (see Results Monitoring Section below).
5. **It is proposed to implement the Project over a period of three years.** The CAT bond will mature in 2.5 years. The 6 additional months will provide sufficient time to calculate payouts and premiums in the event of a long-lingering tropical cyclone hits Jamaica close to the maturity date of the CAT bond³⁰.

Financial Management

6. **Overall financial management risk is expected to be low.** There will be no transfer of Project funds to the GOJ, as the GRiF funded risk premium and transaction costs for the CAT bond will be directly disbursed to Treasury and paid by Treasury to the CAT bond investors and service providers for the risk transfer transaction, providing insurance cover to Jamaica. Therefore, it is expected that no interim financial reports and annual audits are required.

Disbursement arrangements

7. **Upon instructions from the implementing entity, Grant proceeds will be disbursed from the Grant Account to TRE** according to the terms and conditions of the Grant Agreement. TRE then will pay Transaction Costs and Premiums according to the terms and conditions of the Engagement Letter.
8. **The only disbursement method to be used will be advances.** The Engagement Letter will state that advances will be made to a Designated Account managed by TRE. On an annual basis, TRE will report, to the Recipient, the amount of actual payments made for risk premiums and transaction costs. Given the Project design, in order to mitigate TRE's operational risk, and as stated in the EL, there will be a single disbursement of the total Grant proceeds. Until properly documented, by the GOJ to the WB, amounts disbursed into the Designated Account will remain outstanding and, in the event that payments made by TRE to investors, for premiums, and for transaction costs to service providers, are less than total amounts advanced to the Designated Account, TRE will reimburse such balance to the Grant Account, no later than four months after the closing date. The DFIL will describe disbursement arrangements and the TRE-managed account into which advances

³⁰ In the remote event that parameters from the primary Reporting Agency (National Hurricane Centre) cannot be obtained, the CAT bond could be extended by up to 4 months, and in this remote scenario a payout would come later than that set out in the payment timeline. This remote scenario is allowed for in the CAT bond documentation and the RTA. During this extension period, extension premiums would be due to bondholders and we would continue to incur some small transaction costs (Intralinks fees).



will be made. The documentation of Eligible Expenditures will be done using Statements of Expenditure and its minimum frequency will be quarterly.

Table 1: Grant Proceeds Table

Category	Amount of the Grant Allocated (expressed in USD)	Percentage of Expenditures to be Financed (inclusive of Taxes)
(1) Premia	14,850,000	100percent
(2) Transaction Costs*	1,515,000	100percent
TOTAL AMOUNT	16,365,000	

*Estimated amounts. Transactions costs cannot exceed USD 5,000,000. Total Grant amount is fixed.

Procurement

9. **There will be no contracts for goods, works, consulting and non-consulting services subject to WB procurement rules for IPFs, so there are no associated risks.** Any procurement conducted by TRE will be done according to the WB's corporate procurement policies.

Strategy and Approach for Implementation Support

10. The implementation support strategy was defined based on the nature of the proposed Project. After Project approval, the GOJ should request the grant proceeds to be disbursed to TRE (for payment of transaction costs and risk premiums) at which point marketing of the CAT bond to investors can commence. The TRE team, who has broad experience in placing similar risk transfer instruments, will affect this transaction. Moreover, given that the transaction is executed in a highly regulated capital market, the WB will provide strong technical and operational oversight and supervision of the risk transfer transaction. Supervision missions would be conducted as required and virtual monitoring and supervision meetings will be organized as needed. The implementation support strategy will be revisited regularly, considering implementation progress and continuous risk assessment.

Implementation Support Plan and Resource Requirements

Time	Focus	Skills Needed	Resource Estimate (staff weeks/year-SW)
First twelve months	Project Management	Task Team Leader Legal Specialist Actuarial Specialist Financial/Insurance Specialist	8 SWs 6 SWs 6 SWs 6 SWs
	Environmental and Social instruments.	Environmental specialist Social Specialists	2 SWs 2 SWs



12-36 months	Project Management	Task Team Leader Legal Specialist, Actuarial Specialist, Financial/Insurance Specialist.	6 SWs 3 SWs 3 SWs 3SWs
	Environmental and Social standards supervision	Environmental specialist Social specialist	3 SWs 3 SWs



ANNEX 2: Catastrophe Bonds

Catastrophe (CAT) Bonds

CAT bonds allow entities exposed to natural disaster risk, to transfer a portion of that risk to bond investors. CAT bonds work in a similar manner to insurance, paying out when a disaster event meets certain pre-defined criteria (e.g., a specified earthquake magnitude).

In a typical CAT bond structure, the entity exposed to the risk (known as the “sponsor”) enters into an insurance contract with a SPV that issues the Bonds to investors. The SPV invests the proceeds of the bond issuance in highly rated securities that are held in a collateral trust, and it transfers the return on this collateral, together with the insurance premiums received from the sponsor, to the investors as periodic coupons on the Bonds.

If a specified natural disaster occurs during the term of the bond, some or all of the assets held as collateral are liquidated and that money is paid to the sponsor as a pay-out under its insurance contract with the SPV. If no specified event occurs, the collateral assets are liquidated on the maturity date of the Bonds and the money is paid to the investors.

World Bank issued CAT bonds do not require an SPV. Instead, the sponsor (in this case the GOJ) enters into an insurance or derivative contract with the IBRD. The IBRD issues the Bonds to investors, uses the proceeds for development purposes as with other bond issuances, and manages payments to the sponsor and investors.

What is the main difference between CAT bonds and conventional insurance?

CAT bonds allow clients to access a much larger pool of capital (i.e., the trillions of dollars held by bond investors), and in general, longer coverage periods than conventional insurance.

CAT bonds are fully funded transactions (i.e., investors put up all their money upfront by purchasing the Bonds) and therefore there is no risk to the sponsor of default by the investors. In contrast, insurance products involve no upfront payments by the insurer. Rather, the insurer only makes payments if and when a triggering event occurs. Therefore, the client is exposed to the potential default of the insurance provider.



Proposed Structure



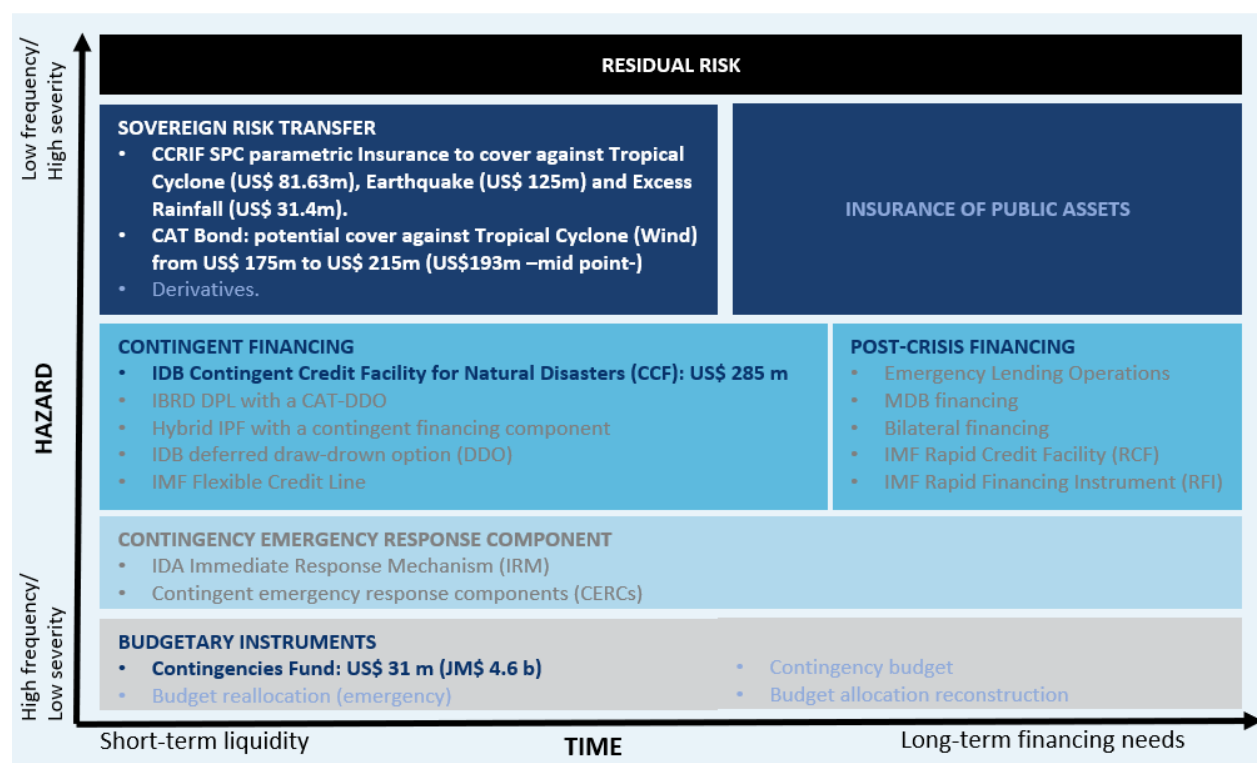


ANNEX 3: Risk Layering Approach

Risk Layering Approach

This approach prioritizes the most cost-effective solution for different layers of risk.

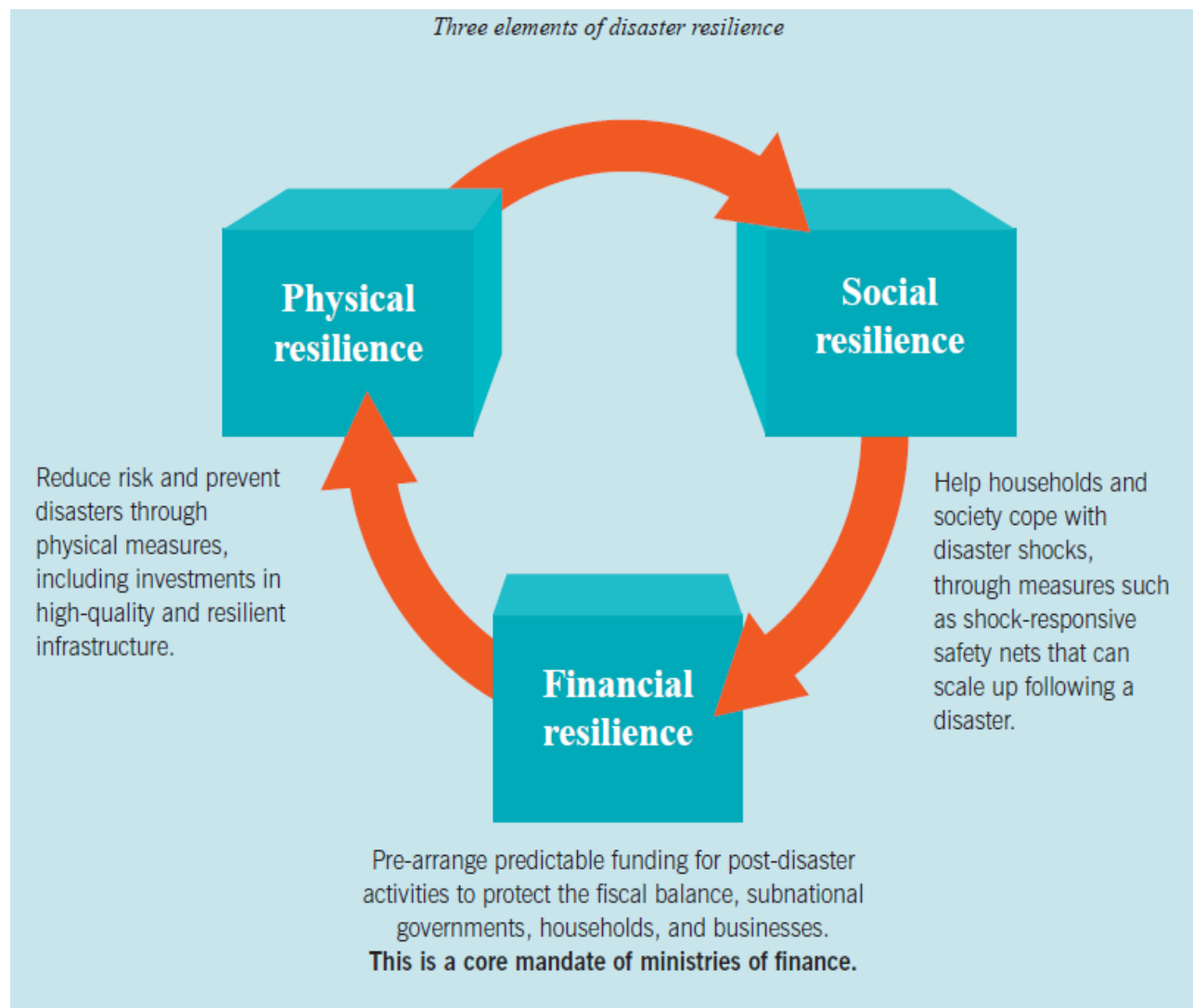
- ✓ **Low-risk layer.** Frequent low-impact events could be financed primarily through risk retention mechanisms in the form of a disaster fund, a dedicated budget line, a contingency budget or budget reallocations. For this layer, the GOJ has currently in place a *Contingencies Fund* whose balance as of February 2021 was US\$31 million.
- ✓ **Medium-risk layer.** Medium-scale, less frequent events could be financed through contingent financing facilities. For this layer, the GOJ has contracted with the IADB a parametric-based *Contingent Credit Facility for Natural Disasters* to protect against tropical cyclones (wind and/or flood) and earthquakes; the credit line is for up to US\$285 million and the payout in case of an event depends on the “affected population”.
- ✓ **High-risk layer.** The financial risk for extreme events that occur infrequently could be transferred to the international capital and reinsurance market using insurance, catastrophe derivatives, or CAT bonds. To manage these risks, the GOJ has purchased parametric insurance to CCRIF SPC to cover against earthquakes (US\$125 million), excess rainfalls (US\$32 million) and tropical cyclones (US\$94 million). The CAT bond fits in this layer.
- ✓ **Residual-risk layer.** Uninsurable risk given the extent of losses in the most catastrophic events.



Source: authors, based on World Bank, 2017. “Sovereign Climate and Disaster Risk Pooling: World Bank Technical Contribution to the G20”.



ANNEX 4: 360 Degree Resilience Framework





List of ongoing reforms supported by the World Bank towards 360-degree resilience

	Financial Resilience
1	Operationalization of the fiscal council
2	Rationalization of the public bodies to reduce fiscal costs
3	Issuance of post-disaster budget execution guidelines for ministries, departments, and agencies to expedite funding for disaster relief and recovery
4	Promulgation of a new Procurement Act that allows the use of appropriate procurement methods for emergency situations
5	Finalization of the National Public Financial Management Policy Framework for Natural Disaster Risk Financing
6	Issuance of new regulations geared toward improving and deepening the local private insurance market to support disaster risk management for households, firms, and government
	Physical Resilience
7	Implementation of the National Building Act of 2018 to ensure resilient construction
8	Finalization of a National Strategy for Coastal Management and Beach Restoration
	Social Resilience
9	Implementation of a community engagement plan expanding the number of eligible beneficiaries in remote areas in the Program of Advancement through Health and Education (PATH), to provide timely support to poor and disaster-affected households.